## II. SIGNIFICANT ACHIEVEMENTS:

### A. Maize Crop Improvement:
### SEASON: Rabi -2022-23

## 1. CO-ORDINATED TRIALS

| | | | | |
|---|---|---|---|---|
| 1. | Experiment 1 (Trial No 1081) | : | Performance of NIVT-Medium maturity maize genotypes for yield | |
| 2. | Objective | : | To identify high yielding maize genotypes in Medium maturity group | |
| 3. | Experimental Details : | | | |
| | a) Location : ZARS, Mandya | | b) Design : Alpha | c) No. of entries: 33+2checks+1Filler =36 |
| | d)Irrigation /Rainfed: Irrigated | | e) Replication : Three | f) Fertilizer (NPK Kg/ha) :150:75:40 |
| | g) Spacing : 60cm x 20 cm | | h) Plot Size: 4.8 $m^2$ | i)Weedicide: Atrazine 2.5kg/ha |
| | j)Zinc Sulpate: 10kg/ha | | k)Date of Sowing:10.12.2022 | l)Date of Harvesting :21-04-2023 |
| 4. | Results: | | | |

| Sl. No. | Genotype | Days to 50% Pollen Shedding | Days to 50% Silking | Days to 75% Dry husk | Plant Height (cm) | Ear Height (cm) | Plant Stand ('000/ha) | Shelling (%) | Yield (t/ha) |
|---|---|---|---|---|---|---|---|---|---|
| 1 | AH 4672 | 64 | 67 | 111 | 214 | 113 | 53.9 | 86.1 | 4.69 |
| 2 | AH 4682 | 69 | 70 | 112 | 214 | 114 | 54.9 | 83.2 | 7.09 |
| 3 | AH 8194 | 74 | 76 | 114 | 207 | 115 | 67.1 | 81.2 | 5.83 |
| 4 | AH 8728 | 70 | 72 | 111 | 213 | 104 | 66.0 | 82.8 | 7.23 |
| 5 | AHD 2065 | 69 | 71 | 112 | 210 | 105 | 75.5 | 81.8 | 11.06 |
| 6 | BAUMH 21-1 | 68 | 69 | 110 | 219 | 116 | 35.1 | 80.9 | 5.44 |
| 7 | BAUMH 21-3 | 69 | 70 | 112 | 200 | 109 | 73.5 | 82.3 | 9.52 |
| 8 | BAUMH 21-4 | 67 | 69 | 111 | 221 | 129 | 62.5 | 81.0 | 6.16 |
| 9 | BH 417167 | 70 | 71 | 112 | 211 | 99 | 78.6 | 82.3 | 9.45 |
| 10 | BRM 19-1 | 74 | 75 | 113 | 215 | 109 | 50.2 | 81.4 | 7.25 |
| 11 | BRM 19-2 | 75 | 75 | 113 | 231 | 108 | 53.3 | 82.3 | 6.16 |
| 12 | BS 652 | 69 | 71 | 112 | 222 | 112 | 77.8 | 81.6 | 12.46 |
| 13 | Ellora Virat-12 | 69 | 71 | 112 | 212 | 106 | 72.0 | 84.3 | 8.71 |
| 14 | IMH 2-22R-1 | 70 | 73 | 111 | 215 | 104 | 69.6 | 80.8 | 8.12 |
| 15 | IMH 2-22R-10 | 71 | 73 | 112 | 218 | 124 | 80.8 | 80.5 | 12.91 |
| 16 | IMH 2-22R-11 | 71 | 73 | 112 | 225 | 116 | 70.9 | 78.4 | 8.43 |
| 17 | IMH 2-22R-12 | 67 | 69 | 113 | 203 | 104 | 71.6 | 87.0 | 9.21 |
| 18 | IMH 2-22R-2 | 66 | 68 | 112 | 194 | 108 | 69.7 | 87.3 | 9.12 |
| 19 | IMH 2-22R-3 | 68 | 69 | 113 | 203 | 99 | 73.5 | 81.9 | 8.64 |
| 20 | IMH 2-22R-4 | 70 | 71 | 110 | 209 | 102 | 78.2 | 80.7 | 10.23 |
| 21 | IMH 2-22R-5 | 70 | 71 | 112 | 218 | 120 | 69.5 | 80.5 | 9.17 |
| 22 | IMH 2-22R-6 | 70 | 71 | 111 | 213 | 110 | 67.4 | 77.6 | 7.85 |
| 23 | IMH 2-22R-7 | 70 | 70 | 112 | 212 | 113 | 64.7 | 82.5 | 9.99 |
| 24 | IMH 2-22R-8 | 70 | 72 | 112 | 223 | 118 | 69.6 | 77.8 | 8.2 |
| 25 | IMH 2-22R-9 | 71 | 73 | 111 | 224 | 110 | 63.0 | 83.0 | 8.25 |
| 26 | KMH 22-5 | 71 | 73 | 112 | 212 | 115 | 65.0 | 81.7 | 10.39 |
| 27 | MMH 22-45 | 69 | 71 | 112 | 216 | 107 | 55.5 | 83.2 | 6.79 |
| 28 | MMH 22-52 | 68 | 70 | 111 | 213 | 114 | 57.3 | 83.5 | 6.21 |
| 29 | OMH 22R-2 | 76 | 77 | 114 | 220 | 103 | 23.3 | 78.1 | 3.33 |
| 30 | SYN 227 705 | 73 | 74 | 113 | 211 | 115 | 79.3 | 84.5 | 11.31 |
| 31 | SYN 227808 | 69 | 71 | 111 | 210 | 106 | 76.2 | 81.8 | 11.14 |
| 32 | SYN 227809 | 66 | 68 | 111 | 202 | 101 | 73.6 | 82.1 | 12.00 |
| 33 | VAH 20001 | 67 | 70 | 111 | 196 | 110 | 70.1 | 83.5 | 6.74 |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 34 | Bio 9544 (check) | . | . | . | GF. | . | . | . | |
| 35 | DHM 117 (check) | 71 | 73 | 112 | 205 | 107 | 62.6 | 82.6 | 9.32 |
| 36 | CMH 08-282 (filler) | 68 | 69 | 111 | 217 | 106 | 65.4 | 79.2 | 8.45 |
| | General mean | 69.6 | 71.3 | 111.7 | 212.8 | 109.9 | 65.7 | 82.0 | 8.481 |
| | | | | | | | | | |
| | CD @ (5 %) | 1.94 | 2.04 | 1.74 | 20.19 | 15.37 | 11.48 | 2.15 | 1.99 |
| | C.V (%) | 1.70 | 1.75 | 0.95 | 5.79 | 8.54 | 12.35 | 1.6 | 14.33 |
| | Note: GF | Germination | failure | | | | | | |

5. **Inference:** Out of thirty six maize hybrids evaluated under medium maturity group along with two checks significant differences were found for grain yield .The hybrid IMH 2-22R-10 (12.91 t/ha) found superior over the check hybrid DHM 117 (9.32 t/ha).
6. **Scientists involved:** P.Mahadevu, Ravindrababu.B.T, Shivakumar.B.S, Mallikarjuna.N, Jadesha.G and Chandrakalahanagi.

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|---|---|---|---|---|---|
| 1. | **Experiment 2 (Trial No.1082 )** | : | **Performance of NIVT-Late duration genotypes for yield** | | |
| 2. | **Objective** | : | **To identify high yielding maize genotypes in latematurity group** | | |
| 3. | **Experimental Details :** | | | | |
| | a) Location : ZARS, Mandya | | b) Design : Alpha | c) No. of entries: 24+3 checks =27 | |
| | d)Irrigation /Rainfed: Irrigated | | e) Replication : Three | f) Fertilizer (NPK Kg/ha) :150:75:40 | |
| | g) Spacing : 60cm x 20 cm | | h) Plot Size: 4.8 $m^2$ | i)Weedicide: Atrazine 2.5kg/ha | |
| | j)Zinc Sulpate: 10kg/ha | | k)Date of Sowing:10.12.2022 | l)Date of Harvesting :24-04-2023 | |
| 4. | **Results:** | | | | |

| Sl. No. | Genotype | Days to 50% Pollen Shedding | Days to 50% Silking | Days to 75% Dry husk | Plant Height (cm) | Ear Height (cm) | Plant Stand ('000/ha) | Shelling (%) | Yield (t/ha) |
|---|---|---|---|---|---|---|---|---|---|
| 1 | ADV 7126 | 75 | 77 | 113 | 209 | 117 | 74.3 | 83.2 | 9.05 |
| 2 | ADV 7251 | 74 | 77 | 115 | 204 | 108 | 68.1 | 85.0 | 8.19 |
| 3 | AH 8245 | GF | | | | | | | |
| 4 | AHD 8721 | 73 | 75 | 114 | 205 | 107 | 68.1 | 79.2 | 7.96 |
| 5 | AHD 8722 | 74 | 77 | 113 | 215 | 105 | 64.6 | 80.7 | 7.22 |
| 6 | BH 417157 | 70 | 72 | 113 | 207 | 106 | 59.0 | 78.7 | 7.61 |
| 7 | BS 650 | 69 | 70 | 113 | 203 | 110 | 72.9 | 80.8 | 8.29 |
| 8 | DM 9445 | 73 | 74 | 114 | 207 | 108 | 69.4 | 82.9 | 9.05 |
| 9 | Ellora Commando 1112 | 72 | 75 | 114 | 206 | 110 | 62.5 | 84.4 | 7.84 |
| 10 | KMH 8333 | 73 | 75 | 113 | 216 | 113 | 72.9 | 85.2 | 9.02 |
| 11 | KMH 8543 | 74 | 76 | 114 | 205 | 111 | 50.0 | 82.2 | 6.70 |
| 12 | KMH 8585 | 72 | 73 | 112 | 206 | 113 | 71.5 | 83.8 | 10.06 |
| 13 | MMH 22-62 | 75 | 77 | 115 | 206 | 108 | 65.3 | 80.0 | 6.59 |
| 14 | MMH 22-67 | 75 | 77 | 115 | 199 | 117 | 37.5 | 81.9 | 7.42 |
| 15 | PM 22201L | 71 | 72 | 110 | 205 | 107 | 73.6 | 81.4 | 10.05 |
| 16 | PM 22202L | 71 | 72 | 113 | 215 | 111 | 70.8 | 80.0 | 10.56 |
| 17 | PM 22203L | 73 | 74 | 114 | 217 | 121 | 72.2 | 82.8 | 9.76 |
| 18 | PM 22204L | 69 | 71 | 111 | 210 | 112 | 67.4 | 84.7 | 10.76 |
| 19 | PM 22205L | 70 | 71 | 112 | 215 | 111 | 76.4 | 85.7 | 10.07 |
| 20 | PM 22206L | 69 | 71 | 114 | 206 | 107 | 73.6 | 84.7 | 9.78 |
| 21 | PM 22207L | 73 | 74 | 112 | 203 | 100 | 69.4 | 83.0 | 9.40 |
| 22 | PM 22208L | 71 | 73 | 112 | 216 | 110 | 71.5 | 82.6 | 9.79 |
| 23 | SMH 6222 | 74 | 76 | 116 | 203 | 108 | 76.4 | 79.8 | 9.79 |
| 24 | SMH 6333 | 71 | 73 | 112 | 209 | 107 | 63.9 | 85.2 | 9.34 |
| 25 | KHM 25K45 (C) | 73 | 75 | 114 | 211 | 109 | 61.1 | 79.8 | 8.78 |
| 26 | NMH 713 (C) | 73 | 75 | 112 | 203 | 110 | 71.5 | 81.3 | 7.43 |
| 27 | P 3522 (C) | 75 | 76 | 113 | 207 | 119 | 70.8 | 81.5 | 8.01 |
| | General Mean | 72.2 | 73.9 | 113 | 207.5 | 110.6 | 66.7 | 82.3 | 8.789 |
| | CD@ (5%) | 4.16 | 4.44 | 2.06 | 15.89 | 10.33 | 10.87 | 2.02 | 2.09 |

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|---|---|---|---|---|---|---|---|---|---|
| | C.V (%) | 3.51 | 3.66 | 1.11 | 4.67 | 5.7 | 9.92 | 1.50 | 14.5 |

5. **Inference:** Out of 27 hybrids evaluated under late maturity group for grain yield. Among the hybrid PM 22204L (8.78 t/ha) recorded maximum grain yield over the best check KHM 25K45 (13.54 t /ha).

6. **Scientists involved:** P.Mahadevu, Ravindrababu.B.T, Shivakumar.B.S, Mallikarjuna.N, Jadesha.G and Chandrakalahanagi

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|---|---|---|---|---|
| 1. | **Experiment 3 (Trial No.1084 )** | : | **Performance of AVT-I Late maturity maize genotypes for yield** | |
| 2. | **Objective** | : | **To identify high yielding maize genotypes in late maturity group** | |
| 3. | **Experimental Details :** | | | |
| | a) Location : ZARS, Mandya | | b) Design : RBD | c) No. of entries:6+3 checks=09 |
| | d)Irrigation/Rainfed: Irrigated | | e) Replication : Three | f) Fertilizer (NPK Kg/ha) :150:75:40 |
| | g) Spacing : 60cm x 20 cm | | h) Plot Size: 9.6m$^2$ | i)Weedicide: Atrazine 2.5kg/ha |
| | j)Zinc Sulpate: 10kg/ha | | k)Date of Sowing:10.12.2022 | l)Date of Harvesting :22-04-2023 |
| 4. | **Results:** | | | |

| Sl. No. | Genotype | Days to 50% Pollen Shedding | Days to 50% Silking | Days to 75% Dry husk | Plant Height (cm) | Ear Height (cm) | Plant Stand ('000/ha) | Shelling (%) | Yield (t/ha) |
|---|---|---|---|---|---|---|---|---|---|
| 1 | ADV 7415 | 70 | 72 | 113 | 216 | 108 | 46.87 | 81.3 | 8.49 |
| 2 | ADV 7559 | 70 | 71 | 116 | 218 | 116 | 29.17 | 80.8 | 7.41 |
| 3 | BH 417193 | 67 | 68 | 112 | 206 | 103 | 71.87 | 79.9 | 13.39 |
| 4 | PM21204L | 69 | 70 | 114 | 237 | 101 | 69.10 | 83.2 | 13.49 |
| 5 | PM21207L | 67 | 67 | 112 | 230 | 97 | 72.92 | 81.8 | 11.86 |
| 6 | TMMH 2845 | 66 | 68 | 114 | 201 | 99 | 70.14 | 79.9 | 11.82 |
| 7 | KMH25K45 (Check) | 70 | 72 | 115 | 217 | 101 | 65.97 | 81.2 | 12.11 |
| 8 | NMH 713 (Check) | 70 | 72 | 115 | 221 | 107 | 59.72 | 81.8 | 9.91 |
| 9 | P 3522 (Check) | 70 | 71 | 112 | 226 | 113 | 73.96 | 81.5 | 11.19 |
| **General mean** | | 68.7 | 70.19 | 113.44 | 219.04 | 105.1 | 62.19 | 81.25 | 11.08 |
| **C D@ (5%)** | | 2.40 | 2.3 | 3.6 | 23 | 14.12 | 14.58 | 3.71 | 3.20 |
| **C V(%)** | | 2.02 | 1.89 | 1.83 | 6.07 | 7.76 | 13.54 | 2.63 | 16.67 |

5. **Inference:** Nine maize hybrids were evaluated under AVT-I late medium maturity group along with three checks. The hybrid PM21204L (13.49 t/ha) found higher grain yield over the best check KMH25K45 hybrids (I2.11 t/ha).

6. **Scientists involved:** P.Mahadevu, Ravindrababu.B.T, Shivakumar.B.S, Mallikarjuna.N., Jadesha.G and Chandrakalahanagi.

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|---|---|---|---|---|---|
| 1. | **Experiment 4 (Trial No.1085 )** | : | **Performance of AVT-II late maturity maize genotypes for yield** | | |
| 2. | **Objective** | : | **To identify high yielding maize genotypes in late maturity group** | | |
| 3. | **Experimental Details :** | | | | |
| | a) Location : ZARS, Mandya | | b) Design : RBD | c) No. of entries:4+3checks=07 | |
| | d)Irrigation /Rainfed: Irrigated | | e) Replication : Three | f) Fertilizer (NPK Kg/ha) :150:75:40 | |
| | g) Spacing : 60cm x 20 cm | | h) Plot Size: 14.4m$^2$ | i)Weedicide: Atrazine 2.5kg/ha | |
| | j)Zinc Sulpate: 10kg/ha | k)Date of Sowing: 17.12.2022 | | l)Date of Harvesting : 28.04.2023 | |
| 4. | **Results:** | | | | |

| Sl. No. | Genotype | Days to 50% Pollen Shedding | Days to 50% Silking | Days to 75% Dry husk | Plant Height (cm) | Ear Height (cm) | Plant Stand ('000/ha) | Shelling (%) | Yield (t/ha) |
|---|---|---|---|---|---|---|---|---|---|
| 1 | ADV 7733 | 72 | 73 | 110 | 250 | 118 | 69.4 | 84.2 | 13.11 |
| 2 | PM20205L | 67 | 68 | 110 | 241 | 115 | 62.0 | 83.9 | 13.47 |
| 3 | TMMH 2882 | 68 | 70 | 108 | 234 | 107 | 62.5 | 78.9 | 11.70 |
| 4 | VNR 4597 | 70 | 72 | 109 | 235 | 115 | 58.8 | 82.9 | 11.93 |
| 5 | KMH25K45 (Check) | 71 | 73 | 110 | 230 | 109 | 66.2 | 79.4 | 14.84 |
| 6 | NMH 713 (Check) | 70 | 72 | 110 | 222 | 106 | 65.3 | 78.6 | 11.90 |
| 7 | P 3522 (Check) | 72 | 73 | 109 | 241 | 113 | 71.3 | 82.5 | 13.12 |
| **General mean** | | 70.19 | 71.52 | 109.4 | 236.2 | 112.0 | 65.1 | 81.46 | 12.87 |
| **C D @ (5 %)** | | 1.47 | 1.62 | 1.45 | 22.25 | 11.66 | 8.31 | 1.17 | 1.591 |
| **C V (%)** | | 1.18 | 1.28 | 0.75 | 5.3 | 5.85 | 7.17 | 0.81 | 6.951 |

5. **Inference:** Seven maize hybrids were evaluated under AVT-II late maturity group along with three checks. None of the test hybrids found superior over the checks hybrids evaluated for grain yield, However among test hybrids PM20205L recorded best yield (13.47 t/ha)

6. **Scientists involved:** P.Mahadevu, Ravindrababu.B.T, Shivakumar.B.S, Mallikarjuna.N, Jadesha.G and Chandrakalahanagi

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|---|---|---|---|---|---|
| 1. | **Experiment 5 (Trial No.1086 )** | : | **Performance of AVT--I-Medium maize genotypes for yield** | | |
| 2. | **Objective** | : | **To identify high yielding maize genotypes in medium maturity group** | | |
| 3. | **Experimental Details :** | | | | |
| | a) Location : ZARS, Mandya | | b) Design : RBD | c) No. of entries: 7+2 checks=9 | |
| | d)Irrigation /Rainfed: Irrigated | | e) Replication : Three | f) Fertilizer (NPK Kg/ha) :150:75:40 | |
| | g) Spacing : 60cm x 20 cm | | h) Plot Size: 9.6m$^2$ | i)Weedicide: Atrazine 2.5kg/ha | |
| | j)Zinc Sulpate: 10kg/ha | k)Date of Sowing: 17.12.2022 | | l)Date of Harvesting : 24.04.2023 | |
| 4. | **Results:** | | | | |

| Sl. No. | Genotype | Days to 50% Pollen Shedding | Days to 50% Silking | Days to 75% Dry husk | Plant Height (cm) | Ear Height (cm) | Plant Stand ('000/ha) | Shelling (%) | Yield (t/ha) |
|---|---|---|---|---|---|---|---|---|---|
| 1 | BH 417182 | 67 | 68 | 111 | 241 | 117 | 67.7 | 78.5 | 13.99 |
| 2 | BH 417206 | 68 | 69 | 110 | 238 | 116 | 74.3 | 78.4 | 14.74 |
| 3 | BMH 18-2 | 67 | 68 | 110 | 234 | 123 | 64.6 | 78.8 | 13.42 |
| 4 | IMHSB 20R-11 | 68 | 69 | 110 | 262 | 139 | 77.1 | 78.6 | 15.40 |
| 5 | IMHSB 20R-3 | 66 | 68 | 110 | 253 | 129 | 78.1 | 83.8 | 14.89 |
| 6 | IMHSB 21R-7 | 69 | 71 | 110 | 237 | 114 | 68.4 | 79.3 | 10.15 |
| 7 | SYN217806 | 67 | 69 | 109 | 251 | 125 | 69.4 | 81.2 | 13.97 |
| 8 | Bio 9544 (C) | GF | | | | | | | |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 9 | DMH 117 (C) | 71 | 73 | 110.33 | 235 | 131 | 49.31 | 77.7 | 8.612 |
| | **General mean** | 67.92 | 69.25 | 109.88 | 243.92 | 124.25 | 68.62 | 79.52 | 13.15 |
| | **C .D@ (5%)** | 1.34 | 1.37 | 1.41 | 14.49 | 17.57 | 12.62 | 1.02 | 2.712 |
| | **CV (%)** | 1.13 | 1.13 | 0.73 | 3.39 | 8.08 | 10.5 | 0.73 | 11.78 |
| | **Note:** | GF :Germination failure | | | | | | | |

5 **Inference:** Out of nine hybrids evaluated under AVT-I medium maturity group for grain yield along with two checks, the hybrid IMHSB 20R-11 (15.40 t/ha) recorded significantly higher grain yield over the best check DMH 117 (8.61 t/ha).

6 **Scientists involved:** P.Mahadevu, Ravindrababu.B.T, Shivakumar.B.S, Mallikarjuna.N, Jadesha.G and Chandrakalahanagi.

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|---|---|---|---|---|---|
| **1.** | **Experiment 6 (Trial No.1087 )** | **:** | **Performance of AVT—II-Medium maize genotypes for yield** | | |
| **2.** | **Objective** | **:** | **To identify high yielding maize genotypes in medium maturity group** | | |
| **3.** | **Experimental Details :** | | | | |
| | a) Location : ZARS, Mandya | | b) Design : RBD | c) No. of entries: 5+2 checks=10 | |
| | d)Irrigation /Rainfed: Irrigated | | e) Replication : Three | f) Fertilizer (NPK Kg/ha) :150:75:40 | |
| | g) Spacing : 60cm x 20 cm | | h) Plot Size: 14.4m$^2$ | i)Weedicide: Atrazine 2.5kg/ha | |
| | j)Zinc Sulpate: 10kg/ha | | k)Date of Sowing: 17.12.2022 | l)Date of Harvesting : 24.04.2023 | |
| **4.** | **Results:** | | | | |

| Sl. No. | Genotype | Days to 50% Pollen Shedding | Days to 50% Silking | Days to 75% Dry husk | Plant Height (cm) | Ear Height (cm) | Plant Stand ('000/ha) | Shelling (%) | Yield (t/ha) |
|---|---|---|---|---|---|---|---|---|---|
| 1 | DKC 8225 | 67 | 69 | 111 | 247 | 131 | 73.15 | 85.7 | 12.11 |
| 2 | IMHSB 20R-6 | 68 | 70 | 113 | 220 | 107 | 77.31 | 82.6 | 10.41 |
| 3 | IMHSB 20R-10 | 67 | 69 | 111 | 219 | 117 | 73.84 | 79.3 | 9.24 |
| 4 | SYN207660 | 68 | 70 | 114 | 216 | 111 | 75.46 | 84.5 | 13.78 |
| 5 | SYN207884 | 69 | 71 | 114 | 226 | 111 | 73.38 | 85.0 | 11.38 |
| 6 | Bio 9544 (Check) | 73 | 74 | 116 | 209 | 112 | 28.47 | 78.9 | 6.02 |
| 7 | DHM 117 (Check) | 70 | 73 | 113 | 229 | 126 | 56.25 | 78.4 | 7.62 |
| | **General mean** | 68.86 | 70.69 | 113.07 | 223.81 | 116.4 | 65.41 | 82.03 | 10.08 |
| | **C D @(5%)** | 2.17 | 3.69 | 2.73 | 19.85 | 11.95 | 7.87 | 0.94 | 2.33 |
| | **CV (%)** | 1.77 | 2.14 | 1.36 | 4.99 | 5.77 | 6.77 | 0.65 | 12.99 |

5 **Inference:** Out of seven hybrids evaluated under AVT-I –II medium maturity group for grain yield along with two checks, the hybrid SYN207660 (13.78 t/ha) recorded significantly higher grain yield over the best check DHM 117 (7.62 t/ha).

6 **Scientists involved:** P.Mahadevu, Ravindrababu.B.T, Shivakumar.B.S, Mallikarjuna.N, Jadesha.G and Chandrakalahanagi

| 1. | Experiment 7 (Trial No1083 ) | : | Performance of Sweet corn maize (I-II-III) genotypes for yield | |
|---|---|---|---|---|
| 2. | Objective | : | To identify high yielding sweet corn maize genotypes | |
| 3. | Experimental Details : | | | |
| | a) Location : ZARS, Mandya | | b) Design : RBD | c) No. of entries: 5+3 checks=08 |
| | d)Irrigation /Rainfed: Irrigated | | e) Replication : Three | f) Fertilizer (NPK Kg/ha) :150:75:40 |
| | g) Spacing : 60cm x 20 cm | | h) Plot Size: 9.6m$^2$ | i)Weedicide: Atrazine 2.5kg/ha |
| | j)Zinc Sulpate: 10kg ha | k)Date of Sowing: 10.12.2022 | | l)Date of Harvesting : 10.03.2023 |
| 4. | Results: | | | |

| Sl. No. | Genotype | Days to 50% Pollen Shedding | Days to 50% Silking | Plant Stand (ha) | TSS (%) | Plant Height (cm) | Ear Height (cm) | Green cob yield without husk (kg/ha) | Green cob yield withhusk (kg/ha) |
|---|---|---|---|---|---|---|---|---|---|
| 1 | BSCH 418184 | 64 | 66 | 60764 | 14.7 | 167 | 77 | 16875 | 20556 |
| 2 | HI BRIX 33 | 70 | 72 | 40972 | 12.6 | 180 | 80 | 12188 | 18021 |
| 3 | HI BRIX 59 | 70 | 72 | 43750 | 14.3 | 176 | 82 | 16319 | 20208 |
| 4 | HI BRIX 81 | 71 | 73 | 54861 | 13.9 | 201 | 102 | 20139 | 25000 |
| 5 | MSCH 22- 28 | 64 | 66 | 41667 | 16.6 | 170 | 89 | 8993 | 11983 |
| 6 | ASKH 4 (Check) | 66 | 68 | 56944 | 16.3 | 200 | 103 | 12014 | 15035 |
| 7 | MISHTI (Check) | 65 | 67 | 62500 | 14.7 | 164 | 75 | 18299 | 21667 |
| 8 | VL Sweet Corn (check) | 63 | 65 | 60764 | 15.5 | 170 | 72 | 10590 | 12431 |
| | General Mean | 66.7 | 68.5 | 52777.8 | 14.8 | 178.5 | 85.0 | 14427.1 | 18112.41 |
| | C.D @(5%) | 1.75 | 2.21 | 12910.4 | 5.98 | 19.16 | 14.6 | 3237.3 | 4429.79 |
| | C.V (%) | 1.5 | 1.84 | 13.97 | 1.55 | 6.13 | 9.8 | 12.81 | 13.97 |

5. **Inference:** Out of eight hybrids evaluated under Sweet corn maize (I-II-III) group for green cob with husk yield along with two checks, the Sweet corn hybrid HI BRIX 81 recorded maximum green cob yield (25000kg/ha) yield over the best check MISHTI (21667kg/ha).

6. **Scientists involved:** P.Mahadevu, Ravindrababu.B.T, Shivakumar.B.S, Mallikarjuna.N, Jadesha.G and Chandrakalahanagi.

| 1. | Experiment 8 (Trial No. 1096 ) | : | Performance of QPM I-II-III Maize genotypes for yield | |
|---|---|---|---|---|
| 2. | Objective | : | To identify high yielding QPM maize genotypes | |
| 3. | Experimental Details : | | | |
| | a) Location : ZARS, Mandya | | b) Design : RBD | c) No. of entries: 7+5checks =12 |
| | d)Irrigation /Rainfed: Irrigated | | e) Replication : Three | f) Fertilizer (NPK Kg/ha) :150:75:40 |
| | g) Spacing : 60cm x 20 cm | | h) Plot Size: 9.6m$^2$ | i) Weedicide: Atrazine 2.5kg/ha |
| | j)Zinc Sulpate: 10kg/ha | k)Date of Sowing: 10.12.2022 | | l)Date of Harvesting : 20.04.2023 |
| 4. | Results: | | | |

| Sl. No. | Genotype | Days to 50% Pollen Shedding | Days to 50% Silking | Days to 75% Dry husk | Plant Height (cm) | Ear Height (cm) | Plant Stand ('000/ha) | Shelling (%) | Yield (t/ha) |
|---|---|---|---|---|---|---|---|---|---|
| 1 | APQH1 | 69 | 71 | 118 | 212 | 104 | 41.7 | 78.6 | 7.07 |
| 2 | APQH5 | 69 | 72 | 115 | 205 | 107 | 46.5 | 75.9 | 6.79 |
| 3 | APQH7 | 69 | 70 | 116 | 215 | 109 | 35.4 | 76.9 | 5.02 |
| 4 | IQH7-215 | 66 | 68 | 111 | 196 | 97 | 56.6 | 80.9 | 6.03 |
| 5 | IQH7-219 | 64 | 66 | 111 | 203 | 105 | 71.5 | 84.4 | 10.72 |
| 6 | MQH 21-51 | 67 | 69 | 115 | 211 | 109 | 54.5 | 80.1 | 8.14 |
| 7 | QPMMH 22-35 | 69 | 70 | 114 | 219 | 111 | 59.0 | 75.8 | 7.33 |
| 8 | HQPM1(Check) | GF | | | | | | | |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 9 | HQPM-5(Check) | 70 | 72 | 114 | 220 | 104 | 59.4 | 75.7 | 6.52 |
| 10 | HQPM-7(Check) | 70 | 72 | 115 | 204 | 100 | 66.3 | 76.7 | 8.25 |
| 11 | LQMH1 (Check) | 62 | 65 | 112 | 192 | 92 | 60.4 | 85.5 | 10.23 |
| 12 | PUSA BIOFORTIFIED MAIZE HYBRID-1 (Check) | 69 | 70 | 111 | 206 | 103 | 60.8 | 76.2 | 7.06 |
| | General Mean | 67.6 | 69.58 | 113.82 | 207.45 | 103.7 | 55.65 | 78.76 | 7.56 |
| | C.D @ (5%) | 2.1 | 2.43 | 2.91 | 20.04 | 11.7 | 12.74 | 1.92 | 1.629 |
| | CV (%) | 1.8 | 2.05 | 1.5 | 5.67 | 6.6 | 13.45 | 1.43 | 12.65 |
| | Note: Germination failure | | | | | | | | |

5. **Inference:** Out of twelve QPM hybrids evaluated for grain yield along with five checks, the hybrid IQH7-219 recorded higher grain yield of (10.72t/ha) over best check LQMH1(10.23t/ha).

6. **Scientists involved:** P.Mahadevu, Ravindrababu.B.T., Shivakumar.B.S., Chandrakalahanagi and Mallikarjuna.N., Jadesha.G.

| | | | | | |
|---|---|---|---|---|---|
| **1.** | **Experiment 9 (Trial No. 1088 )** | **:** | **Evaluation Popcorn maize hybrids for grain yield** | | |
| **2.** | **Objective** | **:** | **To identify high yielding popcorn for grain yield** | | |
| **3.** | **Experimental Details :** | | | | |
| | a) Location : ZARS, Mandya | | b) Design : RBD | c) No. of entries:13+3C=16 | |
| | d)Irrigation /Rainfed: Irrigated | | e) Replication : Three | f) Fertilizer (NPK Kg/ha) :150:75:40 | |
| | g) Spacing : 60cm x 20 cm | | h) Plot Size: 9.6m$^2$ | i)Weedicide: Atrazine 2.5kg/ha | |
| | j)ZincSulpate: 10kg/ha | | k)Date of Sowing: 17.12.2022 | l)Date of Harvesting : 24.04.2023 | |
| **4.** | **Results:** | | | | |

| Sl. No. | Genotype | Days to 50% Pollen Shedding | Days to 50% Silking | Days to 75% Dry husk | Plant Height (cm) | Ear Height (cm) | Plant Stand (ha) | Popping (%) | Shelling (%) | Yield (kg/ha) |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | APCH 4 | 67 | 68 | 110 | 213 | 119 | 66667 | 93.8 | 76.7 | 4628 |
| 2 | CHCP 2204 | 64 | 65 | 107 | 212 | 116 | 63542 | 50.9 | 78.2 | 6857 |
| 3 | CHCP 2205 | 63 | 65 | 107 | 210 | 113 | 67361 | 22.8 | 75.2 | 6687 |
| 4 | CHCP 2206 | 65 | 67 | 108 | 217 | 122 | 65972 | 43.2 | 74.9 | 7183 |
| 5 | CHCP 2207 | GF | | | | | | | | |
| 6 | HPC 4 | 62 | 64 | 109 | 198 | 105 | 44792 | 92.4 | 80.7 | 4226 |
| 7 | HPC 5 | | | | GF | | | | | |
| 8 | IMHP 2101 | 64 | 65 | 108 | 200 | 102 | 69097 | 66.4 | 83.2 | 9415 |
| 9 | IPH 2-22R-1 | 61 | 63 | 107 | 208 | 113 | 74653 | 73.0 | 80.5 | 9196 |
| 10 | MPC 2141 | 65 | 67 | 109 | 190 | 101 | 57639 | 58.9 | 82.2 | 4539 |
| 11 | MPCH 22-22 | 67 | 69 | 110 | 195 | 109 | 47222 | 90.2 | 84.8 | 5054 |
| 12 | PP 09 | 62 | 63 | 105 | 178 | 98 | 70833 | 98.4 | 81.4 | 6670 |
| 13 | PP II | 60 | 62 | 105 | 188 | 95 | 69792 | 97.9 | 80.9 | 6989 |
| 14 | BPCH 6 (Check) | 63 | 65 | 108 | 207 | 115 | 67708 | 66.0 | 80.1 | 5992 |
| 15 | DMRHP 1402(Check) | 62 | 63 | 109 | 183 | 103 | 71528 | 72.9 | 84.9 | 6760 |
| 16 | LPCH 3 (Check) | 63 | 65 | 108 | 195 | 101 | 78472 | 46.8 | 83.2 | 9243 |
| Location Mean | | 63 | 65 | 108 | 200 | 108 | 65377 | 69.5 | 80 | 6674 |
| CD @ (5%) | | | | | | | | | | 1575.025 |
| CV (%) | | | | | | | | | | 13.983 |

5. **Inference:** Out of Sixteen Popcorn hybrids evaluated for grain yield along with three checks, the hybrid popcorn IMHP-2101 (9415 kg/ha) recorded higher grain yield over best check hybrid popcorn LPCH 3 (9243 kg/ha).

6. **Scientists involved:** P.Mahadevu, Ravindrababu.B.T, Shivakumar.B.S, Chandrakalahanagi and Mallikarjuna.N, Jadesha.G.

| | | | |
|---|---|---|---|
| 1. | **Experiment 10 ( Baby Corn ( I-II-III)-1097** | : | **Performance of Baby corn genotypes for baby corn yield** |
| 2. | **Objective** | : | **To identify high yielding Baby corn genotypes** |
| 3. | **Experimental Details :** | | |
| | a) Location : ZARS, Mandya | b) Design : RCBD | c) No. of entries: 7+3check =10 |
| | d)Irrigation /Rainfed: Irrigated | e) Replication : Four | f) Fertilizer (NPK Kg/ha) :150:75:40 |
| | g) Spacing : 60cm x 15 cm | h) Plot Size: 4.8 m² | i) Weedicide: Atrazine 2.5kg/ha |
| | j)Zinc Sulpate: 10kg/ha | k). Date of Sowing:**10**.12.2022 | l). Date of Harvesting : 13/02/2023 to 24-02-2023 |
| 4. | **Results:** | | |

| Sl. No. | Genotype | Days to 50% pollen shedding | Days to 50% Silking | Final Plant (ha) | Ear height (cm) | Plant height (cm) | Length of baby corn (cm) | Diameter of babycorn (cm) |
|---|---|---|---|---|---|---|---|---|
| 1 | BAUMH 21-1-1 | GF | | | | | | |
| 2 | BAUMH 21-8 | 68 | 70 | 86806 | 105 | 213 | 10.27 | 0.61 |
| 3 | IBH 9-224 | 71 | 72 | 86111 | 101 | 182 | 9.30 | 0.59 |
| 4 | IBH9-225 | 71 | 72 | 95833 | 87 | 172 | 9.71 | 0.61 |
| 5 | IIVRBCH 1613 | 65 | 68 | 91667 | 86 | 196 | 9.66 | 0.57 |
| 6 | MBC 21-10 | 72 | 74 | 60417 | 109 | 206 | 8.42 | 0.61 |
| 7 | MBC 22-32 | 70 | 72 | 48611 | 99 | 193 | 8.96 | 0.57 |
| 8 | ABSH-1 (Check) | 66 | 67 | 60417 | 91 | 179 | 9.89 | 0.57 |
| 9 | AH 7043 (Check) | 63 | 67 | 93056 | 73 | 160 | 10.01 | 0.61 |
| 10 | CMVL Baby corn 1 (check) | 64 | 67 | 97222 | 84 | 175 | 10.29 | 0.67 |
| | Location Mean | 67.67 | 69.81 | 80015 | 92.78 | 186.26 | 9.61 | 0.60 |
| | CD@ (5%) | 3.19 | 3.35 | 15273.5 | 11.22 | 17.81 | 1.34 | 0.07 |
| | CV (%) | 2.73 | 2.77 | 11.03 | 6.99 | 5.52 | 8.05 | 7.07 |

| Sl. No. | Genotype | Green Fodder weight ( kg/ha) | First picking days after sowing (days) | Last picking days after sowing (days) | Duration of picking (Days) | Baby corn weight with husk (kg/ha) | Green Ear Weight With husk (kg/ha) |
|---|---|---|---|---|---|---|---|
| 1 | BAUMH 21-1-1 | GF | | | | | |
| 2 | BAUMH 21-8 | 44861 | 70 | 76 | 5 | 7981 | 1355 |
| 3 | IBH 9-224 | 37708 | 70 | 76 | 5 | 8588 | 1467 |
| 4 | IBH9-225 | 36194 | 74 | 78 | 4 | 5455 | 857 |
| 5 | IIVRBCH 1613 | 35694 | 68 | 75 | 7 | 6171 | 1207 |
| 6 | MBC 21-10 | 39028 | 74 | 79 | 5 | 5458 | 903 |
| 7 | MBC 22-32 | 29167 | 72 | 78 | 6 | 5946 | 788 |
| 8 | ABSH-1 (Check) | 29236 | 67 | 73 | 5 | 5975 | 969 |
| 9 | AH 7043 (Check) | 28611 | 67 | 74 | 7 | 5283 | 1467 |
| 10 | CMVL Baby corn 1 (check) | 41250 | 67 | 73 | 7 | 8997 | 1691 |
| | Location Mean | 35750 | 70.0 | 75.7 | 5.7 | 6650.3 | 1189.3 |
| | CD@ (5%) | 8928.5 | 2.93 | 3.91 | 2.81 | 3562.88 | 672.44 |
| | CV (%) | 14.43 | 2.42 | 2.98 | 28.46 | 30.95 | 32.67 |

Note: GF (Germination failure)

5. **Inference:** Out of ten baby corn hybrids evaluated for green cob with husk yield along with three checks none of test hybrid baby corn recorded higher yield over checks. However the among the test baby corn hybrids the best was IBH 9-224 (8558kg/ha) and BAUMH 21-8 (7981kg/ha).

6. **Scientists involved:** P.Mahadevu, Ravindrababu.B.T, Shivakumar.B.S, Chandrakalahanagi and Mallikarjuna.N, Jadesha.G.

## 2. STATION TRIALS:

| | | | | | |
|---|---|---|---|---|---|
| **1.** | **Experiment 11** | **:** | **Evaluation private company maize hybrids for grain yield** | | |
| **2.** | **Objective** | **:** | **To identify high yielding maize hybrids for grain yield** | | |
| **3.** | **Experimental Details :** | | | | |
| | a) Location : ZARS, Mandya | | b) Design : RBD | c) No. of entries:13+2 checks=15 | |
| | d)Irrigation /Rainfed: Irrigated | | e) Replication : Two | f) Fertilizer (NPK Kg/ha) :150:75:40 | |
| | g) Spacing : 60cm x 20 cm | | h) Plot Size: 9.6m$^2$ | i)Weedicide: Atrazine 2.5kg/ha | |
| | j)Zinc Sulpate: 10kg/ha | k)Date of Sowing: 17.12.2022 | | l)Date of Harvesting :26.04.2023 | |
| **4.** | **Results:** | | | | |

| Sl. No. | Hybrids | Days to 50% pollen shedding | Days to 50% silking | Days to 75% dry husk | Plant height (cm) | Ear height (cm) | Shelling % |
|---|---|---|---|---|---|---|---|
| 1 | CP -201 | 71 | 73 | 112 | 231 | 125 | 80.7 |
| 2 | CP -202 | 71 | 73 | 112 | 235 | 126 | 78.9 |
| 3 | Bio-9544 (NC) | 71 | 73 | 112 | 235 | 130 | 80.2 |
| 4 | MAH-14-5 (LC) | 69 | 71 | 110 | 215 | 118 | 78.7 |
| General mean | | 68 | 70 | 112 | 230 | 124 | 78 |

| Sl. No. | Hybrids | No. of rows/ cob | No. of grains/ row | Grain yield (Kg/ha) | % superiority over NC | % superiority Over LC |
|---|---|---|---|---|---|---|
| 1 | CP -201 | 15.4 | 35 | 12260 | 6.7 | 16.0 |
| 2 | CP -202 | 15.0 | 35 | 13654 | 18.9 | 29.2 |
| 3 | Bio-9544 (NC) | 15.2 | 36 | 11486 | | |
| 4 | MAH-14-5 (LC) | 15.3 | 32 | 10568 | | |
| General mean | | 16 | 32 | 11165 | | |
| CD@ (5%) | | | | 1965.7 | | |
| CV (%) | | | - | 10.4 | | |

**5. Inference:** Among two hybrids evaluated with two checks for grain yield, the hybrid CP-202 (13654 kg/ha) recorded significantly higher grain yield followed by CP-201 (12260 kg/ha) over the national check (NC) Bio-9544 (11486 kg/ha) and local check (LC) MAH 14-5 (10568 kg/ha). Further the hybrid CP -202 shown 18.9 and 29.2% superiority, while CP-201 shown 6.7 and 16.0% and superiority over NC and LC respectively.

**6. Scientists involved:** P.Mahadevu and Ravindrababu.B.T

| 1. | **Experiment 12** | : | **Evaluation private company maize hybrids for grain yield** |
|---|---|---|---|
| 2. | **Objective** | : | **To identify high yielding maize hybrids for grain yield** |
| 3. | **Experimental Details :** | | |
| | a) Location : ZARS, Mandya | b) Design : RBD | c) No. of entries:4+2 checks=06 |
| | d)Irrigation /Rainfed: Irrigated | e) Replication : Four | f) Fertilizer (NPK Kg/ha) :150:75:40 |
| | g) Spacing : 60cm x 20 cm | h) Plot Size: 9.6m$^2$ | i)Weedicide: Atrazine 2.5kg/ha |
| | j)Zinc Sulpate: 10kg/ha | k)Date of Sowing: 17.12.2022 | l)Date of Harvesting :26.04.2023 |
| 4. | **Results:** | | |

| Sl. No. | Hybrids | Days to 50% pollen shedding | Days to 50% silking | Days to 75% dry husk | Plant height (cm) | Ear height (cm) | Shelling % |
|---|---|---|---|---|---|---|---|
| 1 | Rank (1M) | 60 | 62 | 106 | 263 | 137 | 79.6 |
| 2 | Bio-9544 (NC) | 60 | 62 | 103 | 256 | 135 | 81.6 |
| 3 | MAH-14-5 (LC) | 59 | 61 | 105 | 261 | 142 | 78.2 |
| General mean | | 59 | 61 | 106 | 256 | 143 | 80 |

| Sl. No. | Hybrids | No. of rows/ cob | No. of grains/ row | Grain yield (Kg/ha) | % superiority over NC | % superiority Over LC |
|---|---|---|---|---|---|---|
| 1 | Rank (1M) | 15 | 36 | 12114 | 21.9 | 17.1 |
| 2 | Bio-9544 (NC) | 16 | 38 | 9935 | | |
| 3 | MAH-14-5 (LC) | 16 | 36 | 10346 | | |
| | General mean | 17 | 35 | 10534 | | |
| | CD@ (5%) | - | - | 1072 | | |
| | CV (%) | - | - | 6.69 | | |

**5. Inference:** Among the hybrids evaluated with two checks for grain yield, the hybrid Rank 1M (12114 kg/ha) recorded significantly higher grain yield over the national check (NC) Bio-9544 (9935 kg/ha) and local check (LC) MAH 14-5 (10346 kg/ha). Further, the hybrid Rank 1M had shown 21.9 and 17.1% superiority, over NC and LC respectively.

**6. Scientists involved:** P.Mahadevu and Ravindrababu.B.T

| 1. | **Experiment 13** | : | **Evaluation of station new maize hybrids for grain yield** |
|---|---|---|---|
| 2. | **Objective** | : | **To identify high yielding maize hybrids for grain yield** |
| 3. | **Experimental Details :** | | |
| | a) Location : ZARS, Mandya | b) Design : RBD | c) No. of entries:94+4 check=98 |
| | d)Irrigation /Rainfed: Irrigated | e) Replication : two | f) Fertilizer (NPK Kg/ha) :150:75:40 |
| | g) Spacing : 60cm x 20 cm | h) Plot Size: 9.6m$^2$ | i)Weedicide: Atrazine 2.5kg/ha |
| | j)Zinc Sulpate: 10kg/ha | k)Date of Sowing: 16.2.2022 | l)Date of Harvesting : 26.06.2023 |
| 4. | **Results:** | | |

| Sl. No. | Hybrids | Days to 50% pollen shedding | Days to 50% silking | Plant height (cm) | Ear height (cm) | Shelling % | Cob length (cm) | Cob girth (cm) | No. of rows / cob | No. of grains / cob | Yield (kg/ha) |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | MAH-22-23 | 60 | 62 | 278 | 155 | 81.3 | 17.6 | 5.08 | 16 | 41 | 19710 |
| 2 | MAH-22-28 | 59 | 61 | 278 | 157 | 77.0 | 18.9 | 5.7 | 16 | 38 | 17430 |
| 3 | MAH-22-17 | 62 | 64 | 275 | 140 | 78.7 | 19.8 | 5.34 | 16 | 40 | 16840 |
| 4 | MAH-22-16 | 64 | 63 | 265 | 147 | 78.1 | 20.54 | 5.28 | 14 | 43 | 16760 |
| 5 | MAH-22-9 | 61 | 64 | 235 | 145 | 76.7 | 17.6 | 5.26 | 14 | 36 | 15670 |
| 6 | MAH-22-33 | 61 | 64 | 275 | 162 | 77.3 | 17.8 | 5.28 | 15 | 40 | 15410 |
| 7 | MAH-22-107 | 63 | 65 | 245 | 140 | 81.3 | 16.58 | 5.4 | 15 | 35 | 14620 |
| 8 | MAH-22-80 | 62 | 65 | 278 | 182 | 80.6 | 19.2 | 4.96 | 15 | 46 | 13710 |
| 9 | MAH-22-32 | 60 | 62 | 248 | 170 | 80.9 | 21.52 | 4.98 | 14 | 44 | 13630 |
| 10 | MAH-22-12 | 60 | 62 | 272 | 153 | 77.1 | 17.2 | 5.6 | 16 | 38 | 13330 |
| 11 | MAH-22-83 | 60 | 62 | 274 | 149 | 83.9 | 17.4 | 4.86 | 15 | 43 | 9260 |
| 12 | MAH-22-21 | 62 | 62 | 275 | 160 | 76.3 | 19.2 | 5.24 | 17 | 41 | 13200 |
| 13 | MAH-22-22 | 58 | 60 | 280 | 172 | 79.5 | 17.3 | 5.16 | 16 | 44 | 12670 |
| 14 | MAH-22-113 | 57 | 60 | 263 | 138 | 79.9 | 17 | 5.18 | 15 | 34 | 10340 |
| 15 | MAH-22-4 | 57 | 59 | 275 | 148 | 76.1 | 20.54 | 5.2 | 15 | 40 | 10060 |
| 16 | MAH-22-11 | 62 | 64 | 268 | 150 | 75.5 | 17.52 | 4.48 | 13 | 42 | 9700 |
| 17 | MAH 19-2 | 62 | 64 | 257 | 140 | 84.2 | 17 | 4.86 | 14 | 40 | 12810 |
| 18 | Bio-9544 | 60 | 62 | 260 | 145 | 86.3 | 16 | 4.22 | 12 | 37 | 6010 |
| 19 | MAH 15-84 | 60 | 62 | 263 | 138 | 77.0 | 17.25 | 5.06 | 17 | 38 | 3690 |
| 20 | MAH 14-5 | 59 | 61 | 270 | 157 | 79.3 | 15.4 | 4.42 | 15 | 37 | 9590 |
| 21 | MAH 14-138 | 61 | 63 | 260 | 172 | 81.9 | 15.4 | 5.02 | 15 | 33 | 11760 |

**5. Inference:** During Khrif 2022, totally 103 new hybrids were developed and evaluation done during Rabi 2022-23, out of 103 hybrids evaluated six hybrids viz, MAH 22-28 (17430 kg/ha), MAH 22-23 (19710 kg/ha), MAH 22-17 (16840 kg/ha), MAH 22-16 (16760 kg/ha) and MAH 22-09 (15670 kg/ha) recorded good yield compared to local check hybrid MAH 14-5 (95900 kg/ha) and national check Bio 9544 (60100 kg/ha). These hybrids were further evaluated for cob characters and growth parameters.

**6. Scientists involved:** P.Mahadevu and Shivakumar.B.S.

| | | | |
|---|---|---|---|
| 1. | **Experiment 14** | : | **Evaluation of station MLT maize hybrids for grain yield** |
| 2. | **Objective** | : | **To identify high yielding maize hybrids for grain yield** |
| 3. | **Experimental Details :** | | |
| | a) Location : ZARS, Mandya | b) Design : RBD | c) No. of entries:4+2 check = 6 |
| | d)Irrigation /Rainfed: Irrigated | e) Replication : Three | f) Fertilizer (NPK Kg/ha) :150:75:40 |
| | g) Spacing : 60cm x 20 cm | h) Plot Size: 9.6m$^2$ | i)Weedicide: Atrazine 2.5kg/ha |
| | j)Zinc Sulpate: 10kg/ha | k)Date of Sowing: 17.12.2022 | l)Date of Harvesting :26.04.2023 |
| 4. | **Results:** | | |

| Sl. No | Hybrids | Days to 50% flowering | Days to 50% Silking | Day s to 75 % dry husk | Plant height (cm) | Ear height (cm) | Shelling (%) | No. of rows/ Cob | No. of grains /row | Cob length (cm) | cob girth (cm) | Test weight (gm) | Grain yield (Kg/ha) | (%) Superioty over (NC) | (%) Superioty over (LC) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | MAH-19-2 | 68 | 69 | 111 | 229 | 126 | 78.0 | 15.3 | 33 | 18.2 | 5.1 | 35.5 | 11346 | -1.2 | 7.4 |
| 2 | MAH-15-84 | 70 | 72 | 114 | 247 | 133 | 78.5 | 15.7 | 35 | 17.1 | 5.1 | 38.5 | 12240 | 6.6 | 15.8 |
| 3 | MAH-22-46 | 74 | 75 | 116 | 216 | 119 | 79.0 | 15.4 | 26 | 16.5 | 4.8 | 34.0 | 10032 | -12.7 | -5.1 |
| 4 | MAH-22-3 | 74 | 75 | 114 | 206 | 118 | 75.8 | 14.5 | 31 | 17.0 | 4.6 | 36.0 | 9067 | -21.1 | -14.2 |
| 5 | MAH-22-55 | 73 | 75 | 113 | 236 | 125 | 76.7 | 16.2 | 29 | 20.6 | 5.2 | 41.0 | 10229 | -10.9 | -3.2 |
| 6 | **MAH-22-17** | **73** | **75** | **116** | **240** | **124** | **76.6** | **16.0** | **31** | **20.5** | **5.3** | **37.0** | **15130** | **31.7** | **43.2** |
| 7 | MAH-22-113 | 71 | 73 | 115 | 219 | 115 | 77.4 | 14.6 | 33 | 19.1 | 5.1 | 42.0 | 10655 | -7.2 | 0.8 |
| 8 | MAH-21-501 | 71 | 73 | 110 | 222 | 106 | 80.2 | 13.0 | 40 | 19.8 | 4.8 | 35.0 | 10836 | -5.6 | 2.5 |
| 9 | NAH-2049 - I | 66 | 67 | 110 | 235 | 118 | 76.7 | 15.8 | 33 | 17.7 | 5.2 | 38.5 | 10207 | -11.1 | -3.4 |
| 10 | NAH-2049 -II | 67 | 69 | 109 | 242 | 126 | 79.1 | 15.6 | 31 | 17.8 | 5.0 | 40.0 | 9723 | -15.3 | -8.0 |
| 11 | MAH-14-138 | 67 | 70 | 110 | 247 | 141 | 78.6 | 16.4 | 33 | 18.3 | 5.3 | 37.5 | 11492 | 0.1 | 8.7 |
| 12 | BIO-9544 (NC) | 71 | 73 | 112 | 235 | 130 | 80.2 | 15.2 | 36 | 19.1 | 5.0 | 34.5 | 11486 | | 8.7 |
| 13 | MAH-14-5 (LC) | 69 | 71 | 110 | 215 | 118 | 78.7 | 15.3 | 32 | 15.2 | 4.6 | 38.0 | 10568 | -8.0 | |
| | General Mean | 70 | 72 | 112 | 230 | 124 | 78 | 16 | 32 | 18.1 | 5.0 | 37.2 | 11165 | | |
| | | | | | | | | | | | | C.D @ 5 % | 1862.5 | | |
| | | | | | | | | | | | | C.V (%) | 9.987 | | |

**5. Inference:** In this experiment, eight station maize hybrids were evaluated out of which hybrids MAH-22-17 (15130 kg/ha) and MAH-14-138 (11492 kg/ha) were recorded higher yield compared to check hybrids Bio-9544 (11486 kg/ha), MAH-14-5 (10568kg/ha).

**6. Scientists involved:** P.Mahadevu and Ravindrababu.B.T